



Gas battery turns emissions into electricity

Developed by a team at Sungkyunkwan University in collaboration with Ajou University and Chungbuk National University, a 'gas battery' converts greenhouse gases into electricity while simultaneously capturing harmful emissions. The Gas Capture and Electricity Generator (GCEG)



The Gas Capture and Electricity Generator (GCEG) turns emissions into energy (Credit: <https://interestingengineering.com>)

produces direct current power from carbon dioxide and nitrogen oxides without requiring an external energy source, offering a lower-energy alternative to conventional carbon capture systems. The device uses carbon-based electrodes and hydrogel materials to trigger charge redistribution during gas adsorption. The technology could support self-powered environmental sensors, battery-free IoT devices, and industrial energy recovery systems.

Growing ultra-thin magnetic materials on wafers

Researchers at the Indian Institute of Science have developed a wafer-scale technique for growing ultra-thin magnetic materials, a breakthrough that could accelerate spintronics and low-power electronics. The method produces continuous, defect-controlled two-dimensional magnetic films across large wafers, overcoming previous limits of microscopic-scale production. Using



IISc team grows ultra-thin magnetic material at wafer scale (Credit: www.msn.com)

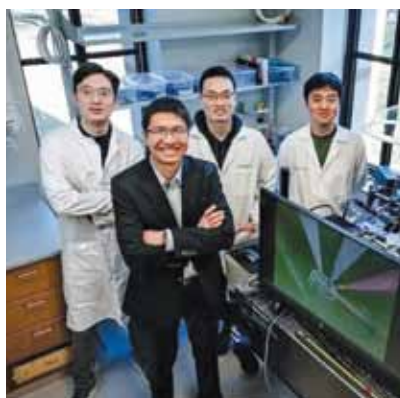
physical vapour transport deposition on synthetic mica substrates, the team created high-quality magnetic layers compatible with semiconductor manufacturing. The films retain their magnetic properties even at atomic thickness, making them promising for data storage, sensors, and advanced computing.

Wearable sweat patch displays biomarkers in real time

Engineered by a multidisciplinary team in Barcelona, Spain, the patch continuously measures key electrolytes and metabolites during exercise and extended activity. Researchers have developed a wearable sweat-monitoring patch that displays biomarkers in real time, enabling continuous tracking of hydration, fatigue, and physical exertion without invasive testing. The flexible device integrates silicon-based sensors, microfluidic channels, and electrochemical detection to analyse sweat directly on the skin. Researchers said the system maintained stable monitoring for over an hour, addressing a major limitation of earlier sweat-sensing wearables.

Microwave-assisted, multi-material 3D printing for embedded electronics

Embedded electronics can now be printed directly into complex structures without damaging surrounding materi-



Professor Yong Lin Kong (front row) with co-authors Xin Yang, Jian Teng and Yu Liu (back row, L to R) (Credit: <https://news.rice.edu>)

als, thanks to a new microwave-assisted 3D printing technique developed at Rice University. The system, called Meta-NFS, uses highly focused microwave energy to selectively heat conductive inks with precision as fine as a human hair, improving multi-material electronics manu-

Chip uses vibrations for efficient power conversion

Engineered by a team at the University of California, San Diego, a chip uses mechanical vibrations instead of traditional magnetic components to improve power conversion efficiency in high-performance electronics. The prototype achieved up to 96.2% efficiency while converting 48V to 4.8V, a key requirement for data centres and AI computing systems. The design relies on piezoelectric resonators to reduce size, heat, and energy losses in power delivery circuits. The technology could help lower electricity consumption in GPU-heavy servers and future compact electronics.